INDENG242A - Individual Proposal

Runpeng Zhu

3043177634

For this project, I am planning to build a personalized restaurant recommendation system that combines a user's historical behavior, such as views, clicks, and purchases, with their current stated preferences like cuisine type, price range, distance, and desired restaurant rating. Using these features together with information about the candidate restaurants, the model would be able to generate a score for all candidate restaurants and use these scores to recommend the top-k restaurants. I plan to begin with a relatively simple gradient tree model as a baseline and then explore more advanced tree based methods, and see how it improves over time. This problem interests me because restaurant choice is highly personalized, and a user's current needs may differ from their historical behavior, so combining long-term behavioral information with current preferences can give better recommendations. The system could also benefit users by reducing the time and effort required to search through many restaurants, and this can also help a restaurant platform to provide more relevant recommendations. I think a successful project should achieve a strong predictive and ranking performance using evaluation metrics like AUC, Precision@K, Recall@K, and Hit Rate@K, and also produce recommendations that are practical and user friendly.